

### Handshake Game

2PP: Two parties A & B.

A sends  $A, N_A$  (nonce of A)

B sends back  $N_A, N_B, MAC_k(N_A, N_B, B \rightarrow A, 2)$

A then sends  $MAC_k(N_A, N_B, A \rightarrow B, 3)$  Nonce is randomly generated for each interaction, counter protects against replay

Note: hash functions and their keys are always known, attackers can freely compute

Don't encrypt within a MAC, because the information can be leaked

### Modular Arithmetic

$$\left(2048 - \frac{8}{1436} + \frac{10785}{20*17}\right) \mod 99$$

$$20 * 17 = 340$$

$$1436^{-1} \mod 99 = 2$$

$$340^{-1} \mod 99 = 76$$

$$(2048 \mod 99) - (8 * 1436^{-1} \mod 99) + (10785 * 340^{-1} \mod 99)$$

$$(68) - (8 * 2 \mod 99) + (10875 * 76 \mod 99)$$

$$68 - (16 \mod 99) + (39 \mod 99) = (68 - 16 + 39) \mod 99$$

$$91 \mod 99 = 91$$

$$b = a^{-1} \mod m, a * b \mod 1 = 1$$

### Euler totient

$$l > 0$$

$$\Phi(p^l) = p^l - p^{l-1} \leftarrow \text{want things in this form}$$

$$\Phi(A * B) = \Phi(A)\Phi(B)$$

$$\Phi(80) = \Phi(2^4 * 5), \text{ Prime Factorization}$$

$$\Phi(80) + \Phi(2729) = \Phi(2^4 * 5) + \Phi(2^3 * 11 * 31)$$

$$\Phi(2^4) * \Phi(5) + \Phi(2^3) * \Phi(11) * \Phi(31)$$

$$(2^4 - 2^3) * (5^1 - 5^0) + (2^3 - 2^2) * (11^1 - 11^0) * (31^1 - 31^0)$$

$$(16 - 8)(5 - 1) + (8 - 4)(11 - 1)(31 - 1) = (8 * 4) + (4 * 10 * 30)$$

### Homomorphism (ElGamal)

The multiplication of two ciphertexts is equal to the encryption of the multiplication of its plaintexts